

Hardware-Portable Quantum Reservoir Computing for Regime-Sensitive VIX Volatility Forecasting

FraQTech – Track A: Financial Volatility Prediction

Abstract

We study whether a frozen quantum dynamical feature map can provide a selective forecasting benefit on regime-sensitive financial volatility while remaining executable on near-term hardware. Earlier pilot simulations across SPY, QQQ, and VIX found the clearest quantum-reservoir result on VIX: a continuous-variable QRC with a temporal convolutional readout attained NRMSE 0.4127 ± 0.0062 , compared with 0.4167 ± 0.0020 for a classical TCN. Phase 3 replaces the weak legacy gate model with a validation-selected temporal Ising QRC, benchmarks it against GARCH, HAR-RV, ESN, ridge, and TCN, and characterizes scaling over 5/10/15 qubits, finite shots, depolarizing and amplitude-damping noise, and physical QPU execution. We additionally implement the common MNIST expressivity benchmark and provide a fully reproducible qBraid workflow. Our claim is deliberately conditional: we test for a task-specific quantum benefit, not universal quantum advantage.

1 Problem, data, and contributions

Volatility forecasting is difficult because market dynamics are nonlinear, persistent, heavy-tailed, and subject to abrupt regime changes. Reservoir computing is a natural match: a fixed nonlinear dynamical system transforms an input history into a high-dimensional state trajectory, while only a lightweight readout is fitted [9]. QRC replaces the classical reservoir with driven quantum dynamics [1, 2]. Recent work has demonstrated few-atom feedback reservoirs, 108-qubit analog reservoirs, robust chaotic forecasting, and direct realized-volatility applications [3–6]. A recent unifying view emphasizes tuning the memory–nonlinearity balance rather than maximizing circuit complexity indiscriminately [8].

Task. From public daily VIX OHLC data (2015–2025), we compute log returns and a five-day rolling volatility proxy v_t . Given the previous 50 trading days, the primary task predicts v_{t+1} . A secondary task predicts whether $t + 1$ lies above a causal rolling 80th-percentile high-volatility threshold, including explicit low-to-high transition recall. Splits are chronological 60/20/20; robust scaling is fitted on training data only. Architecture and hyperparameters are chosen by validation QLIKE, and the test set is evaluated once.

Contributions. (i) A hardware-portable, zero-trainable-quantum-parameter temporal Ising QRC with repeated angle encoding, fixed many-body dynamics, temporal multiplexing, and Z/ZZ observables. (ii) A finance-appropriate comparison against GARCH(1,1), HAR-RV, ESN, ridge-lag, persistence, and TCN using RMSE, NRMSE, MAE, QLIKE, Mincer–Zarnowitz calibration, and prediction-level Diebold–Mariano tests [10–13]. (iii) Reproducible 5/10/15-qubit, shot, noise, QPU, and MNIST studies on qBraid/IBM.

2 Temporal Ising quantum reservoir

For compressed temporal input u_t , each checkpoint applies data re-uploading followed by fixed non-commuting fields and Ising interactions,

$$U_t = \prod_{v=1}^V U_{\text{res}} U_{\text{enc}}(u_t), \quad (1)$$

$$U_{\text{enc}}(u_t) = \prod_i R_z(a_i^z u_t + b_i^z) R_y(a_i^y u_t + b_i^y), \quad (2)$$

$$U_{\text{res}} = \prod_{\ell=1}^L \left[\prod_{(i,j) \in E} R_{zz}(J_{ij}^\ell) \prod_i R_z(h_{i,z}^\ell) R_x(h_{i,x}^\ell) \right]. \quad (3)$$



Figure 1: End-to-end hybrid workflow. Only the classical readout is fitted. The same fixed circuit generator targets exact simulation, Qiskit Aer, IBM Runtime, and qBraid-managed gate QPUs.

Table 1: Frozen Phase 3 evaluation matrix.

Block	Configurations	Outputs
Classical	persistence, GARCH, HAR-RV, ridge, ESN, TCN; 5 seeds	NRMSE, QLIKE, MZ, runtime
QRC scaling	5/10/15 qubits; ridge and TCN; 3 seeds	accuracy/resource curves
Measurement/noise	256/1024/4096 shots; depolarizing and damping; 3 seeds	degradation and feature drift
Hardware	IBM 5q targeted test subset; optional second qBraid gate QPU	depth, gates, shots, jobs, metrics
Regimes	raw, ESN, and QRC features	macro-F1, AUROC, Brier, transition recall
MNIST	fixed 2000/500 subset; 5/10/15 qubits	accuracy, macro-F1, runtime

All angles are fixed by a reported seed. The reservoir state is never variationally trained. At each virtual node we measure all $\langle Z_i \rangle$ and topology-edge $\langle Z_i Z_j \rangle$, plus four global summaries. Ridge is the primary readout because it most directly tests linear accessibility of reservoir information; a compact TCN is a secondary hybrid readout.

Selection and controls. On validation only, a 5-qubit search varies topology, 4/6/8 temporal bins, 1/2 virtual nodes, 1/2 reservoir layers, re-uploading, interaction scale, and ridge penalty. We then freeze the winning design before testing. Controls include Z-only features, no re-uploading, line topology, ESN states, raw windows, and a matched-dimensional nonlinear random projection. These distinguish a useful quantum trajectory from a large arbitrary feature expansion.

3 Experimental protocol and platform use

Table 1 summarizes the final matrix. Exact statevectors establish an ideal reference; Qiskit Aer adds 256/1024/4096-shot sampling and explicit depolarizing/amplitude-damping channels. For hardware, a shallower six-checkpoint, five-qubit ring reservoir is transpiled to the selected backend. IBM EstimatorV2 directly evaluates the Z/ZZ operator set, with optimization level 3, resilience level 1, and XY4 dynamical decoupling. Every result records logical/transpiled depth, two-qubit gates, shots, runtime, backend, and job IDs. An optional qBraid-managed second gate QPU provides cross-platform validation. We do not use annealing hardware because the computation is driven temporal dynamics rather than QUBO optimization.

The common MNIST benchmark resizes digits to 8×8 and interprets rows as an eight-step sequence through the same reservoir family, followed by multinomial logistic regression. This is an expressivity check, not evidence for financial quantum advantage.

4 Results and analysis plan

The prior VIX pilot provides the starting point in Table 2. It motivates VIX as the primary case but does not establish a qubit-hardware benefit because the winning encoder was a continuous-variable moment simulator. The new gate-QRC must therefore be evaluated independently.

The principal statistical question is whether the selected gate-QRC improves QLIKE or NRMSE relative to ESN/ridge/TCN across seeds and whether any improvement survives matched-dimensional random features. Prediction-level Diebold–Mariano tests assess loss differences. Noise results will report both forecast degradation and feature drift from the exact statevector reference. Hardware is interpreted through the chain exact $\rightarrow$ finite-shot $\rightarrow$ noisy simulator $\rightarrow$ QPU; a competitive or gracefully degraded QPU result is meaningful even if it does not beat the best classical TCN.

Table 2: Completed pilot evidence and Phase 3 result slots. New values are generated from raw JSON and prediction CSV artifacts.

Model	Qubits	VIX NRMSE ↓	Status
Classical TCN	0	0.4167 ± 0.0020	completed pilot
CV-QRC + TCN	modes	0.4127 ± 0.0062	completed pilot
Temporal Ising QRC + ridge	5	PENDING	pending execution
Temporal Ising QRC + ridge	10	PENDING	pending execution
Temporal Ising QRC + ridge	15	PENDING	pending execution
IBM QPU subset	5	PENDING	pending execution

5 Stakeholder value, limitations, and reproducibility

A useful volatility forecast can support risk limits, market-making inventory, derivatives hedging, and regime-aware allocation. The practical contribution is not merely a small accuracy change: it is a transparent decision boundary describing when a fixed quantum feature service is worth its circuit and shot cost.

Limitations are explicit. Exact simulation scales exponentially; the QPU test is a targeted subset; market data and volatility proxies are imperfect; the TCN may account for much of the pilot gain; and one dataset cannot support a universal advantage claim. The final claim will therefore be one of *selective quantum benefit* only if gate-QRC gains survive ablations, noise, and fair classical comparison.

The submission includes project-relative code, a frozen data manifest/checksum, one-command smoke tests, prediction-level outputs, qBraid/IBM adapters, a qBraid Skill, and a complete README. Judges can reproduce simulator headlines without credentials and can re-run QPU jobs after configuring their own authorized account. Generative AI was used for code/documentation assistance; all executed results and scientific claims are produced and verified by the team.

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